

| Subject Name                                  | Subject Type        | Activities Planned                                                                                                                                                                                                                                                                                  | Remarks                                                                                                                                                                                                                                 |
|-----------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Structures</b>                        | <b>Theory + Lab</b> | <ul style="list-style-type: none"> <li>30 Practice problems + 15 Final test (Industry specific)</li> <li>Codethon (Industry specific)</li> <li>Gate Questions Practice</li> </ul>                                                                                                                   | NPTEL – Data Structures and Algorithms<br>Codethon and Gate Questions Practice are mapped with Assignment 1 and 2 marks<br>Practice and Final test are mapped with Lab internal marks                                                   |
| <b>Java Programming</b>                       | <b>Theory + Lab</b> | <ul style="list-style-type: none"> <li>30 Practice problems + 15 Final test (Industry specific)</li> <li>Codethon (Industry specific)</li> <li>Mini project</li> </ul>                                                                                                                              | Oracle Academy – Java Foundations<br>Mini project and Codethon are mapped with Assignment 1 and 2 marks<br>Practice and Final test are mapped with Lab internal marks                                                                   |
| <b>Exploratory Data Analysis</b>              | <b>Theory + Lab</b> | Microsoft Data Analysis and Visualization with Power BI <ul style="list-style-type: none"> <li>Module 1: Creating reports, M2: Navigation and Accessibility (12hrs)</li> <li>M3: Bringing Data to the User, M4: Identifying Patterns and Trends (12hrs)</li> <li>M5: Mini Project (3hrs)</li> </ul> | EDA using Tableau<br>Microsoft Power BI Data Analyst Certification<br>First two activities are mapped with Assignment 1 and 2 marks<br>Mini Project is mapped with Lab internal exams                                                   |
| <b>Operating Systems</b>                      | <b>Theory + Lab</b> | <ul style="list-style-type: none"> <li>CSE Gate Question Practice</li> <li>Assessment on Industry specific Questions</li> </ul>                                                                                                                                                                     | Both activities are mapped with Assignment 1 and 2 marks                                                                                                                                                                                |
| <b>Responsive Web Development with React-</b> | <b>Theory + Lab</b> | <ul style="list-style-type: none"> <li>Mini project (Student Portfolio)</li> <li>Internal SIH participation</li> </ul>                                                                                                                                                                              | Version Control (Git+GitHub)<br>Deployment (Vercel/Netlify)<br>Meta Front-End Developer Professional Certificate<br>Both activities are mapped with Assignment 1 and 2 marks<br>Mini Projects are mandatory for Hackathon participation |
| <b>Algorithms</b>                             | <b>Theory</b>       | <ul style="list-style-type: none"> <li>30 Practice problems + one Final assessment test (Industry specific)</li> <li>Case Study / Mini project</li> </ul>                                                                                                                                           | Algorithms Specialization from Stanford University on Coursera<br>Both activities are mapped with Assignment 1 and 2 marks                                                                                                              |
| <b>Database Management Systems</b>            | <b>Theory + Lab</b> | <ul style="list-style-type: none"> <li>30 Practice problems + 15 Final test (Industry specific)</li> <li>Gate Question Practice</li> <li>Mini project</li> </ul>                                                                                                                                    | freeCodeCamp – Relational Database Certification<br>Mini project and Gate Question Practice are mapped with Assignment 1 and 2 marks<br>Practice and Final test are mapped with Lab internal marks                                      |
| <b>Artificial Intelligence Essentials</b>     | <b>Theory</b>       | <ul style="list-style-type: none"> <li>Machine Learning with Scikit-learn, PyTorch &amp; Hugging Face - Coursera</li> <li>Gate Questions Practice</li> </ul>                                                                                                                                        | Microsoft Certified Azure AI Fundamentals (ai-900)<br>AWS Certified AI Practitioner                                                                                                                                                     |

|                                             |                   |                                                                                                                 |                                                          |
|---------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|                                             |                   |                                                                                                                 | Both activities are mapped with Assignment 1 and 2 marks |
| <b>Standards in Artificial Intelligence</b> | <b>Theory</b>     | <ul style="list-style-type: none"><li>● Quiz</li><li>● Internal Hackathon Participation</li></ul>               | Both activities are mapped with Assignment 1 and 2 marks |
| <b>Data Privacy and Security</b>            | <b>Theory</b>     | <ul style="list-style-type: none"><li>● Case study</li><li>● NPTEL: Design Thinking Primer</li></ul>            | Both activities are mapped with Assignment 1 and 2 marks |
| <b>Back-End Development and APIs</b>        | <b>Skill Dev.</b> | <ul style="list-style-type: none"><li>● Mini project</li><li>● National level Hackathon participation</li></ul> | Both activities are mapped with Assignment 1 and 2 marks |